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Exercise Sheet 2

This is not the official exercise sheet from the course.
More difficult exercises are denoted by ★ to ★★★

> If you can solve it, it is an exercise;
> otherwise it's a research problem.
>
> Richard Bellman

Exercise 1.
Prove the following statements using a direct proof (sometimes with case distinction)

1. If $n \in \mathbb{N}$, then $n^2 - n$ is even.
2. If $n \in \mathbb{N}$, then $1 + (-1)^n(2n - 1)$ is a multiple of 4.
3. Let $x, y \in \mathbb{Z}$ be odd, then $x^2 + y^2$ is even but not divisible by 4.
4. Let $n \in \mathbb{Z}$ be odd, then $n^2 - 1$ is a multiple of 8.
5. For any $n \in \mathbb{Z}$, n^2 is either a multiple of 3, or one more than a multiple of 3.
6. Every odd integer is the difference of two squares
7. ★ Let $n \in \mathbb{Z}$. If n is not divisible by 3, then $n^2 + 2$ is divisible by 3.
8. ★ Prove that every prime $p > 3$ is of the form $6n \pm 1$ for some integer n.

Solution 1.
1. We can write

$$n^2 - n = n(n - 1)$$

Since n and $n - 1$ are consecutive integers, exactly one of them must be even. Therefore, their product $n(n - 1)$ is an even number.

2. We proceed by case distinction on the parity of n.

Case 1: n is even. Let $n = 2k$ for some $k \in \mathbb{N}$.

$$\begin{aligned} 1 + (-1)^{2k}(2(2k) - 1) &= 1 + 1 \cdot (4k - 1) \\ &= 4k \end{aligned}$$

which is a multiple of 4.

Case 2: n is odd. Let $n = 2k + 1$ for some $k \in \mathbb{Z}_{\geq 0}$.

$$\begin{aligned} 1 + (-1)^{2k+1}(2(2k+1) - 1) &= 1 - 1 \cdot (4k + 2 - 1) \\ &= 1 - (4k + 1) \\ &= -4k \end{aligned}$$

Since $-4k = 4(-k)$, this is also a multiple of 4. Thus, in all cases, the expression is a multiple of 4.

3. Since x and y are odd, we can write $x = 2k + 1$ and $y = 2m + 1$ for some $k, m \in \mathbb{Z}$.

$$\begin{aligned} x^2 + y^2 &= (2k+1)^2 + (2m+1)^2 \\ &= 4k^2 + 4k + 1 + 4m^2 + 4m + 1 \\ &= 4(k^2 + k + m^2 + m) + 2 \end{aligned}$$

This can be factored as $2(2(k^2+k+m^2+m)+1)$, which shows it is even. However, when divided by 4, the term $4(k^2 + k + m^2 + m)$ leaves a remainder of 0, leaving an overall remainder of 2. Therefore, $x^2 + y^2$ is not divisible by 4.

4. Let $n = 2k + 1$ for some $k \in \mathbb{Z}$.

$$\begin{aligned} n^2 - 1 &= (2k+1)^2 - 1 \\ &= 4k^2 + 4k + 1 - 1 \\ &= 4k(k+1) \end{aligned}$$

Because k and $k + 1$ are consecutive integers, one of them must be even. Thus, their product is even, meaning $k(k + 1) = 2m$ for some $m \in \mathbb{Z}$. Substituting this back yields:

$$n^2 - 1 = 4(2m) = 8m$$

which is a multiple of 8.

5. By the division algorithm, any integer n can be written as $n = 3k$, $n = 3k + 1$, or $n = 3k + 2$ for some $k \in \mathbb{Z}$. We test all three cases:

Case 1: $n = 3k$.

$$n^2 = (3k)^2 = 9k^2 = 3(3k^2)$$

which is a multiple of 3.

Case 2: $n = 3k + 1$.

$$n^2 = (3k+1)^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1$$

which is one more than a multiple of 3.

Case 3: $n = 3k + 2$.

$$\begin{aligned} n^2 &= (3k+2)^2 = 9k^2 + 12k + 4 \\ &= 9k^2 + 12k + 3 + 1 \\ &= 3(3k^2 + 4k + 1) + 1 \end{aligned}$$

which is also one more than a multiple of 3.

6. Let $n = 2k + 1$ for some $k \in \mathbb{Z}$ be an arbitrary odd integer. Consider the squares of the consecutive integers $(k + 1)$ and k. Taking their difference yields:

$$\begin{aligned}(k+1)^2 - k^2 &= (k^2 + 2k + 1) - k^2 \\ &= 2k + 1 \\ &= n\end{aligned}$$

Thus, every odd integer can be written as the difference of two squares.

7.* If n is not divisible by 3, then by the division algorithm, n must be of the form $3k+1$ or $3k + 2$ for some $k \in \mathbb{Z}$.

Case 1: $n = 3k + 1$.

$$\begin{aligned}n^2 + 2 &= (3k + 1)^2 + 2 \\ &= 9k^2 + 6k + 1 + 2 \\ &= 3(3k^2 + 2k + 1)\end{aligned}$$

which is divisible by 3.

Case 2: $n = 3k + 2$.

$$\begin{aligned}n^2 + 2 &= (3k + 2)^2 + 2 \\ &= 9k^2 + 12k + 4 + 2 \\ &= 9k^2 + 12k + 6 \\ &= 3(3k^2 + 4k + 2)\end{aligned}$$

which is also divisible by 3.

8.* By the division algorithm, any integer p can be expressed in one of the following forms for some $k \in \mathbb{Z}$: $6k, 6k + 1, 6k + 2, 6k + 3, 6k + 4$, or $6k + 5$.

Assume $p > 3$ is a prime number. It cannot be divisible by 2 or 3. However:

- $6k = 2(3k)$ is divisible by 2 and 3.
- $6k + 2 = 2(3k + 1)$ is divisible by 2.
- $6k + 3 = 3(2k + 1)$ is divisible by 3.
- $6k + 4 = 2(3k + 2)$ is divisible by 2.

This leaves only $6k + 1$ and $6k + 5$ as possible forms for a prime $p > 3$. Furthermore, we can rewrite $6k + 5$ as:

$$\begin{aligned}6k + 5 &= 6k + 6 - 1 \\ &= 6(k + 1) - 1 \\ &= 6n - 1\end{aligned}$$

where $n = k + 1$. Therefore, every prime $p > 3$ must be of the form $6n + 1$ or $6n - 1$, which can be written concisely as $6n \pm 1$.

Exercise 2.
Prove the following statements using an indirect proof.

1. Let $x \in \mathbb{Z}$. If $x^2 - 5x + 6$ is odd then x is even.
2. Let $r \in \mathbb{R}$. If r^2 is irrational, then r is also irrational.
3. Let $x, y \in \mathbb{R}$. If $x + y \geq 20$, then $x \geq 10$ or $y \geq 10$.
4. Let $a, b \in \mathbb{Z}$. If $a^2 + b^2$ is divisible by 4, then a and b are both even.
5. ★ Let $a, b \in \mathbb{Z}$. If $a^2 + b^2$ is a perfect square, then at least one of a or b is even.
6. ★ Let $n \in \mathbb{N}$ with $n > 1$. If n is a composite number, then n has a prime factor p such that $p \leq \sqrt{n}$.
7. ★★ Let $k \in \mathbb{N}$. If $2^k + 1$ is prime then either $k = 0$ or $k = 2^n$ for some $n \in \mathbb{N}$.

Solution 2.
1.

Proof. Let x be odd. Then we can write $x = 2k + 1$ for some $k \in \mathbb{Z}$. So

$$\begin{aligned} x^2 - 5x + 6 &= (2k+1)^2 - 5(2k+1) + 6 \\ &= 4k^2 + 4k + 1 - 10k - 5 + 6 \\ &= 4k^2 - 6k + 2 \\ &= 2(2k^2 - 3k + 1) \end{aligned}$$

So $x^2 - 5x + 6$ is even.

By indirection, the statement is true. □

2.

Proof. Suppose r is rational. Then we can write $r = \frac{a}{b}$ for some integers a, b with $b \neq 0$. Squaring this gives

$$r^2 = \left(\frac{a}{b}\right)^2 = \frac{a^2}{b^2}$$

Since $a^2, b^2 \in \mathbb{Z}$ and $b^2 \neq 0$, it follows that r^2 is rational.

By contraposition, if r^2 is irrational, then r is also irrational. □

3.

Proof. Suppose that $x < 10$ and $y < 10$. Adding these two inequalities yields

$$x + y < 10 + 10 = 20$$

Thus, $x + y < 20$.

By contraposition, if $x + y \geq 20$, then $x \geq 10$ or $y \geq 10$. □

4.

Proof. Suppose that a and b are not both even. There are three cases to consider:

Case 1: a is even, b is odd. Let $a = 2k$ and $b = 2l + 1$. Then

$$\begin{aligned} a^2 + b^2 &= (2k)^2 + (2l+1)^2 \\ &= 4k^2 + 4l^2 + 4l + 1 \\ &= 2(2k^2 + 2l^2 + 2l) + 1 \equiv 1 \pmod 4 \end{aligned}$$

Case 2: a is odd, b is even. Let $a = 2k + 1$ and $b = 2l$. Then

$$\begin{aligned} a^2 + b^2 &= (2k+1)^2 + (2l)^2 \\ &= 4k^2 + 4k + 1 + 4l^2 \\ &= 2(2k^2 + 2k + 2l^2) + 1 \equiv 1 \pmod 4 \end{aligned}$$

Case 3: Both a and b are odd. Let $a = 2k + 1$ and $b = 2l + 1$. Then

$$\begin{aligned} a^2 + b^2 &= (2k+1)^2 + (2l+1)^2 \\ &= 4k^2 + 4k + 1 + 4l^2 + 4l + 1 \\ &= 4(k^2 + k + l^2 + l) + 2 \equiv 2 \pmod 4 \end{aligned}$$

In all cases, $a^2 + b^2$ is not divisible by 4 (since it leaves a remainder of 1 or 2).

By contraposition, if $a^2 + b^2$ is divisible by 4, then a and b are both even. □

5.

Proof. Suppose that a and b are both odd. Then we can write $a = 2k + 1$ and $b = 2l + 1$ for some $k, l \in \mathbb{Z}$. Then

$$\begin{aligned} a^2 + b^2 &= (2k+1)^2 + (2l+1)^2 \\ &= 4k^2 + 4k + 1 + 4l^2 + 4l + 1 \\ &= 4(k^2 + k + l^2 + l) + 2 \\ &= 2(2(k^2 + k + l^2 + l) + 1) \end{aligned}$$

This shows that $a^2 + b^2 \equiv 2 \pmod 4$. However, the square of any integer is congruent to either 0 or 1 $\pmod 4$, meaning a perfect square can never be congruent to 2 $\pmod 4$. Therefore, $a^2 + b^2$ cannot be a perfect square.

By contraposition, if $a^2 + b^2$ is a perfect square, at least one of a or b must be even. □

6.

Proof. We prove the contrapositive: If n has no prime factor $p \leq \sqrt{n}$, then n is prime.

Suppose that n has no prime factor $p \leq \sqrt{n}$. To show that n is prime, we assume for the sake of contradiction that n is composite. By definition, a composite number can be factored as $n = ab$ for some integers a, b where $1 < a \leq b < n$.

If both $a > \sqrt{n}$ and $b > \sqrt{n}$, then their product would be

$$n = ab > \sqrt{n} \cdot \sqrt{n} = n$$

which implies $n > n$, a contradiction. Therefore, at least one of the factors, say a, must satisfy $a \leq \sqrt{n}$.

Since $a > 1$, by the Fundamental Theorem of Arithmetic, a has a prime factor p (where $p \leq a$). Because p divides a and a divides n, p is also a prime factor of n. Furthermore, we have

$$p \leq a \leq \sqrt{n}$$

This means n has a prime factor less than or equal to $\sqrt{n}$, which directly contradicts our initial assumption.

Therefore, n must be prime. □

7.

Proof. Suppose that $k \in \mathbb{N}$ is such that $k \neq 0$ and k is not a power of 2 (i.e., $k \neq 2^n$ for any $n \in \mathbb{N}$). Then k must have an odd prime factor $d > 1$, allowing us to write $k = m \cdot d$ for some odd integer $d > 1$ and integer $m \geq 0$. Then

$$2^k + 1 = (2^m)^d + 1$$

Since d is odd, we can use the sum of odd powers factorization formula

$$(2^m)^d + 1 = (2^m + 1)\left((2^m)^{d-1} - (2^m)^{d-2} + \cdots + 1\right)$$

Since $d > 1$, both factors on the right-hand side are strictly greater than 1, meaning $2^k + 1$ is composite.

By contraposition, if $2^k + 1$ is prime, then either $k = 0$ or $k = 2^n$ for some $n \in \mathbb{N}$. □

Exercise 3.
Prove the following statements using a proof by contradiction.

1. The number $\sqrt{3}$ is irrational.
2. There are infinitely many prime numbers.
3. Prove that there are no integers x and y such that $6x + 9y = 100$.
4. Prove that there is no smallest positive rational number.
5. Let q be rational and r irrational. Prove that $q + r$ is irrational.
6. ★ Prove that there are no integers a and b such that $a^2 - 4b = 2$.
7. ★ Let $a, b, c \in \mathbb{Z}$. If $a^2 + b^2 = c^2$, prove that at least one of a or b is divisible by 3.

Exercise 4.
Prove the following statements by mathematical induction

1. For $n \in \mathbb{N}$, $\sum_{i=1}^{n}(8i - 5) = 4n^2 - n$.
2. For all $n \in \mathbb{N}$, $3^n > n^2$.
3. For all $n \in \mathbb{N}$, $n^3 - n$ is divisible by 3.
4. ★ Let F_n be the n-th Fibonacci number ($F_1 = 1, F_2 = 1, F_{n+2} = F_{n+1} + F_n$).
   Prove for all $n \in \mathbb{N}$, $\displaystyle\sum_{k=1}^{n} F_k = F_{n+2} - 1$.

Solution 4.
1.

Proof. Suppose for sake of contradiction that $\sqrt{3}$ is rational. Then we can write $\sqrt{3} = \frac{a}{b}$ for some integers a, b with $b \neq 0$ and $\gcd(a, b) = 1$. Squaring this gives

$$\begin{aligned} 3 = \left(\frac{a}{b}\right)^2 &= \frac{a^2}{b^2} \\ a^2 &= 3b^2 \end{aligned}$$

Since $b^2 \in \mathbb{Z}$, a^2 is a multiple of 3. Because 3 is prime, this implies that a is also a multiple of 3, so we can write $a = 3k$ for some integer k. Substituting this back into our equation yields

$$\begin{aligned} (3k)^2 &= 3b^2 \\ 9k^2 &= 3b^2 \\ b^2 &= 3k^2 \end{aligned}$$

This implies b^2 is a multiple of 3, meaning b must also be a multiple of 3. Since both a and b are divisible by 3, they are not coprime, which contradicts our initial assumption that $\gcd(a, b) = 1$.

By contradiction, the statement is true. □

2.

Proof. Suppose for sake of contradiction that there are finitely many prime numbers. Let us list them all as $p_1, p_2, \ldots, p_k$. Consider the integer N defined as

$$N = p_1 p_2 \cdots p_k + 1$$

Since N is strictly greater than any prime p_i in our list, N must either be a new prime number itself, or it must be composite and thus divisible by at least one prime. However, dividing N by any of the primes p_i from our finite list leaves a remainder of 1. Therefore, N is not divisible by any known prime, meaning it must have a prime factor not included in our list. This contradicts the assumption that $p_1, p_2, \ldots, p_k$ represents all prime numbers.

By contradiction, the statement is true. □

3.

Proof. Suppose for sake of contradiction that there exist integers x and y such that $6x + 9y = 100$. We can factor out a 3 on the left side

$$3(2x + 3y) = 100$$

Since x and y are integers, $(2x + 3y)$ is also an integer. This implies that the left side is a multiple of 3, which means that 100 must be divisible by 3. However, $100 = 3 \cdot 33 + 1$, meaning it leaves a remainder of 1 when divided by 3. This is a contradiction.

By contradiction, the statement is true. □

4.

Proof. Suppose for sake of contradiction that there is a smallest positive rational number, which we will call r. Since r is rational and positive, $r = \frac{a}{b}$ for some positive integers $a > 0$ and $b > 0$. Consider the number r' defined as half of r

$$r' = \frac{r}{2} = \frac{a}{2b}$$

Since a and $2b$ are both positive integers, r' is a valid positive rational number. Furthermore, since $r > 0$, it follows algebraically that $0 < r' < r$. This contradicts the assumption that r is the absolutely smallest positive rational number.

By contradiction, the statement is true. □

5.

Proof. Suppose for sake of contradiction that the sum $s = q + r$ is a rational number. Since q is defined as rational, we can write $q = \frac{a}{b}$ and $s = \frac{c}{d}$ for some integers a, b, c, d with $b \neq 0$ and $d \neq 0$. We can solve for r

$$\begin{aligned} r &= s - q \\ &= \frac{c}{d} - \frac{a}{b} \\ &= \frac{bc - ad}{bd} \end{aligned}$$

Since a, b, c, d are all integers and $bd \neq 0$, the resulting fraction $\frac{bc-ad}{bd}$ is a rational number. This implies that r is rational, which directly contradicts the premise that r is irrational.

By contradiction, the statement is true. □

6.

Proof. Suppose for sake of contradiction that there exist integers a and b such that $a^2 - 4b = 2$. Rearranging the equation yields

$$\begin{aligned} a^2 &= 4b + 2 \\ a^2 &= 2(2b + 1) \end{aligned}$$

Since $2b+1$ is an integer, a^2 is a multiple of 2 (an even number). The square of an integer is even if and only if the integer itself is even, so we can write $a = 2k$ for some integer k. Substituting $2k$ for a gives

$$\begin{aligned} (2k)^2 &= 2(2b + 1) \\ 4k^2 &= 4b + 2 \\ 2k^2 &= 2b + 1 \end{aligned}$$

The left side of the equation, $2k^2$, is an even integer because it is a multiple of 2. The right side, $2b+1$, is an odd integer. An even integer cannot equal an odd integer, resulting in a contradiction.

By contradiction, the statement is true. □

7.

Proof. Suppose for sake of contradiction that $a^2 + b^2 = c^2$, but neither a nor b is divisible by 3. By the division algorithm, any integer not divisible by 3 must leave a remainder of 1 or 2 when divided by 3, meaning it can be written as $3k + 1$ or $3k + 2$. Squaring these possible forms yields

$$\begin{aligned} (3k + 1)^2 &= 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1 \equiv 1 \pmod 3 \\ (3k + 2)^2 &= 9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1 \equiv 1 \pmod 3 \end{aligned}$$

Therefore, the square of any integer not divisible by 3 is always congruent to 1 (mod 3). Since neither a nor b is divisible by 3, we have $a^2 \equiv 1 \pmod 3$ and $b^2 \equiv 1 \pmod 3$. Adding these together gives

$$a^2 + b^2 \equiv 1 + 1 \equiv 2 \pmod 3$$

However, we assumed $a^2 + b^2 = c^2$. The integer c^2 is a perfect square, and as shown above, perfect squares can only be congruent to 0 (mod 3) (if $3|c$) or 1 (mod 3) (if $3 \nmid c$). It is impossible for a perfect square to be congruent to 2 (mod 3), which is a contradiction.

By contradiction, the statement is true. □

Solution 4.
1.

Proof. To do a proof by induction, let $P(n)$ be defined by $P(n) = 1$ if and only if $\sum_{i=1}^{n}(8i - 5) = 4n^2 - n$. Step 1 is to prove $P(1)$; this holds trivially because the sum consists of the single term $8(1) - 5 = 3$ and we also have $4(1)^2 - 1 = 4 - 1 = 3$.

Step 2 is to prove that for an arbitrary n, under the assumption $P(n)$, also $P(n + 1)$ is true

$$\begin{aligned}\sum_{i=1}^{n+1}(8i - 5) &= \sum_{i=1}^{n}(8i - 5) + 8(n + 1) - 5 \\ &= (4n^2 - n) + 8n + 3 \\ &= 4n^2 + 7n + 3 \\ &= 4(n^2 + 2n + 1) - n - 1 \\ &= 4(n + 1)^2 - (n + 1).\end{aligned}$$

This concludes the proof of $\forall n P(n)$. □

2.

Proof. To do a proof by induction, let $P(n)$ be defined by $P(n) = 1$ if and only if $3^n > n^2$. Step 1 is to prove $P(1)$; this holds trivially because $3^1 = 3$ and we also have $1^2 = 1$, yielding $3 > 1$. Step 2 is to prove that for an arbitrary n, under the assumption $P(n)$, i.e., $3^n > n^2$, also $P(n + 1)$ is true, i.e., $3^{n+1} > (n + 1)^2$

$$\begin{aligned}3^{n+1} &= 3 \cdot 3^n \\ &> 3 \cdot n^2 \\ &= n^2 + 2n^2\end{aligned}$$

Since $n \in \mathbb{N}$, for $n \geq 1$ we know $2n^2 \geq 2n \geq 2n + 1$ holds for any $n \geq 2$. (For the isolated edge case $n = 1$, $P(2)$ holds directly as $3^{1+1} = 9 > (1 + 1)^2 = 4$). Thus, substituting $2n^2 \geq 2n + 1$ for all necessary cases yields

$$\begin{aligned}n^2 + 2n^2 &\geq n^2 + 2n + 1 \\ &= (n + 1)^2.\end{aligned}$$

This concludes the proof of $\forall n P(n)$. □

3.

Proof. To do a proof by induction, let $P(n)$ be defined by $P(n) = 1$ if and only if $n^3 - n = 3k$ for some integer k. Step 1 is to prove $P(1)$; this holds trivially because $1^3 - 1 = 0$, and $0 = 3(0)$, which is divisible by 3. Step 2 is to prove that for an arbitrary n, under the assumption $P(n)$, i.e., $n^3 - n = 3k$, also $P(n+1)$ is true, i.e., $(n+1)^3 - (n+1) = 3m$ for some integer m

$$\begin{aligned}(n+1)^3 - (n+1) &= (n^3 + 3n^2 + 3n + 1) - n - 1 \\ &= (n^3 - n) + 3n^2 + 3n \\ &= 3k + 3n^2 + 3n \\ &= 3(k + n^2 + n).\end{aligned}$$

Since $(k+n^2+n)$ is an integer, the expression is a multiple of 3. This concludes the proof of $\forall n P(n)$. □

4.

Proof. To do a proof by induction, let $P(n)$ be defined by $P(n) = 1$ if and only if $\sum_{k=1}^{n} F_k = F_{n+2} - 1$. Step 1 is to prove $P(1)$; this holds trivially because the sum consists of the single term $F_1 = 1$, and evaluating the right side utilizing the Fibonacci definition gives $F_{1+2} - 1 = F_3 - 1 = (F_2 + F_1) - 1 = 2 - 1 = 1$. Step 2 is to prove that for an arbitrary n, under the assumption $P(n)$, i.e., $\sum_{k=1}^{n} F_k = F_{n+2} - 1$, also $P(n+1)$ is true, i.e., $\sum_{k=1}^{n+1} F_k = F_{n+3} - 1$

$$\begin{aligned}\sum_{k=1}^{n+1} F_k &= \sum_{k=1}^{n} F_k + F_{n+1} \\ &= (F_{n+2} - 1) + F_{n+1} \\ &= (F_{n+2} + F_{n+1}) - 1 \\ &= F_{n+3} - 1.\end{aligned}$$

This concludes the proof of $\forall n P(n)$. □

Exercise 5.
Prove the following statements using the Pigeonhole Principle.

1. Show that in a group of 13 people, at least two were born in the same month.
2. Show that out of 5 random integers, there are two, whose difference is divisible by 4.
3. A round-robin tournament has 10 players. Prove that at any point during the tournament, there are two players who have played the exact same number of matches so far.
4. ★★★ Let $m, n \in \mathbb{N}$. Prove that any sequence of $mn+1$ distinct real numbers contains either a strictly increasing subsequence of length $m+1$ or a strictly decreasing subsequence of length $n+1$.

Solution 5.
1.

Proof. Consider all months in a year. Then create 12 sets each containing one unique month of the year. Now we distribute the 13 people in to the sets according to their month of birth. Since $13 > 12$, the pigeonhole principle guarantees that at least 2 persons must end up in the same set. This directly implies that at least two were born in the same month □

2.

Proof. Consider the remainder of the integers after dividing by 4. Create 4 different sets each containing the remainder. Now if we distribute the 5 integers into those sets then by the pigeonhole principle, at least two must end up in the same set. This means that they have the same remainder mod 4. So by definition their difference is divisible by 4. □

3.

Proof. In a tournament with $n = 10$ players, the number of matches any individual player has played belongs to the set $\{0, 1, 2, \ldots, n-1\}$, which contains n possible values.

However, a player who has played 0 matches and a player who has played $n-1$ matches cannot co-exist simultaneously: if a player has played 0 matches, then no player could have played against all other $n-1$ players.

Therefore, at any point in the tournament, the set of actual match counts realized by the players is a subset of either $\{0, 1, \ldots, n-2\}$ or $\{1, 2, \ldots, n-1\}$. In both cases, there are at most $n-1$ possible match counts.

Distributing n players into at most $n-1$ possible match counts, the Pigeonhole Principle guarantees that at least two players have played the exact same number of matches. □

4.

Proof. Let $(n_1, n_2, \ldots, n_N)$ be a sequence of $N = mn+1$ distinct real numbers. Label each term n_i with an ordered pair (a_i, b_i), where a_i is the length of the longest monotonically increasing subsequence ending at n_i, and b_i is the length of the longest monotonically decreasing subsequence ending at n_i.

To show that every term receives a distinct label pair, consider any two indices $i < j$:

If $n_i < n_j$, any increasing sequence ending at n_i can be extended by n_j, so $a_i < a_j$.

If $n_i > n_j$, any decreasing sequence ending at n_i can be extended by n_j, so $b_i < b_j$.

Thus, $(a_i, b_i) \neq (a_j, b_j)$ whenever $i \neq j$, giving N distinct pairs of labels.

Suppose for sake of contradiction that there is no increasing subsequence of length $m+1$ and no decreasing subsequence of length $n+1$. So we have $a_i \leq m$ and $b_i \leq n$. Since $a_i, b_i \geq 1$, there are only mn possible label pairs.

By the Pigeonhole Principle, placing $N = mn + 1$ distinct label pairs into mn possible pairs is impossible. Therefore, $\exists k \in \mathbb{N} \setminus \{0\}$ such that $a_k \geq m + 1$ or $b_k \geq n + 1$.

If $a_k \geq m + 1$, then by definition, the longest increasing subsequence ending at n_k has length at least $m + 1$, which guarantees an increasing subsequence of length $m + 1$. Similarly, if $b_k \geq n + 1$, the longest decreasing subsequence ending at n_k has length at least $n + 1$, which guarantees a decreasing subsequence of length $n + 1$.[1] □

[1]See: Seidenberg, A. (1959), "A simple proof of a theorem of Erdős and Szekeres"

Exercise 6.
Simplify the following propositional logic formula.

$$F = \neg\Big((P \to Q) \to (Q \to P)\Big) \vee \Big(P \wedge (P \to Q)\Big)$$

Solution 6.

$$\begin{aligned}
F &\equiv \neg\Big(\neg(\neg P \vee Q) \vee (\neg Q \vee P)\Big) \vee \Big(P \wedge (\neg P \vee Q)\Big) && \text{(Implication)}\\
&\equiv \Big((P \to Q) \wedge \neg(Q \to P)\Big) \vee \Big((P \wedge \neg P) \vee (P \wedge Q)\Big) && \text{(De Morgan, Distributivity)}\\
&\equiv \Big((\neg P \vee Q) \wedge (Q \wedge \neg P)\Big) \vee \Big(\bot \vee (P \wedge Q)\Big) && \text{(Implication, De Morgan, } F \wedge \neg F \equiv \bot)\\
&\equiv (\neg P \wedge Q) \vee (P \wedge Q) && \text{(Absorption, } F \vee \bot \equiv F)\\
&\equiv (\neg P \vee P) \wedge Q && \text{(Distributivity)}\\
&\equiv \top \wedge Q && (F \vee \neg F \equiv \top)\\
&\equiv Q && (F \vee \top \equiv F)
\end{aligned}$$

Exercise 7.
Simplify the following propositional logic formula.

$$F = \Big((P \leftrightarrow Q) \wedge (P \to \neg Q)\Big) \to \neg P$$

Solution 7.

$$\begin{aligned}
F &\equiv \Big(((P \wedge Q) \vee (\neg P \wedge \neg Q)) \wedge (\neg P \vee \neg Q)\Big) \to \neg P && \text{(Equivalence, Implication)}\\
&\equiv \Big(((P \wedge Q) \vee (\neg P \wedge \neg Q)) \wedge \neg(P \wedge Q)\Big) \to \neg P && \text{(De Morgan)}\\
&\equiv \Big(((P \wedge Q) \wedge \neg(P \wedge Q)) \vee ((\neg P \wedge \neg Q) \wedge \neg(P \wedge Q))\Big) \to \neg P && \text{(Distributivity)}\\
&\equiv \Big(\bot \vee ((\neg P \wedge \neg Q) \wedge (\neg P \vee \neg Q))\Big) \to \neg P && (F \wedge \neg F \equiv \bot, \text{ De Morgan})\\
&\equiv (\neg P \wedge \neg Q) \to \neg P && (\bot \vee F \equiv F, \text{ Absorption})\\
&\equiv \neg(\neg P \wedge \neg Q) \vee \neg P && \text{(Implication)}\\
&\equiv (P \vee Q) \vee \neg P && \text{(De Morgan, Double Negation)}\\
&\equiv (P \vee \neg P) \vee Q && \text{(Associativity, Commutativity)}\\
&\equiv \top \vee Q && (F \vee \neg F \equiv \top)\\
&\equiv \top && (\top \vee F \equiv \top)
\end{aligned}$$

Exercise 8.
Simplify the following formula such that it contains at most one $\neg$, one $\vee$, and one $\wedge$.

$$F = \Big((A \to B) \wedge (B \to \neg A)\Big) \vee \neg\Big(A \to \neg(B \wedge C)\Big)$$

Solution 8.

$$\begin{aligned}
F &\equiv \Big((\neg A \vee B) \wedge (\neg B \vee \neg A)\Big) \vee \neg\Big(\neg A \vee \neg(B \wedge C)\Big) && \text{(Implication)} \\
&\equiv \Big(\neg A \vee (B \wedge \neg B)\Big) \vee \Big(A \wedge (B \wedge C)\Big) && \text{(Distributivity, De Morgan)} \\
&\equiv (\neg A \vee \bot) \vee (A \wedge B \wedge C) && (F \wedge \neg F \equiv \bot, \text{ Associativity}) \\
&\equiv \neg A \vee (A \wedge (B \wedge C)) && (F \vee \bot \equiv F) \\
&\equiv (\neg A \vee A) \wedge (\neg A \vee (B \wedge C)) && \text{(Distributivity)} \\
&\equiv \top \wedge (\neg A \vee (B \wedge C)) && (F \vee \neg F \equiv \top) \\
&\equiv \neg A \vee (B \wedge C) && (\top \wedge F \equiv F)
\end{aligned}$$

Exercise 9.
Simplify the following formula such that it contains at most one $\to$ operator and no other logical operators,

$$F = \neg\Big(\neg(P \wedge Q) \to (P \to \neg R)\Big) \vee (Q \to R)$$

Solution 9.

$$\begin{aligned}
F &\equiv \Big(\neg(P \wedge Q) \wedge \neg(P \to \neg R)\Big) \vee (\neg Q \vee R) && \text{(Negation of Implication, Implication)} \\
&\equiv \Big(\neg(P \wedge Q) \wedge (P \wedge R)\Big) \vee (\neg Q \vee R) && \text{(Negation of Implication)} \\
&\equiv \Big((\neg P \vee \neg Q) \wedge P \wedge R\Big) \vee (\neg Q \vee R) && \text{(De Morgan)} \\
&\equiv \Big(((\neg P \wedge P) \vee (\neg Q \wedge P)) \wedge R\Big) \vee (\neg Q \vee R) && \text{(Distributivity)} \\
&\equiv \Big((\bot \vee (P \wedge \neg Q)) \wedge R\Big) \vee (\neg Q \vee R) && (F \wedge \neg F \equiv \bot) \\
&\equiv (P \wedge \neg Q \wedge R) \vee \neg Q \vee R && (\bot \vee F \equiv F, \text{ Associativity}) \\
&\equiv \neg Q \vee R \vee (P \wedge \neg Q \wedge R) && \text{(Commutativity)} \\
&\equiv \neg Q \vee R && \text{(Absorption)} \\
&\equiv Q \to R && \text{(Implication)}
\end{aligned}$$

Exercise 10.
Simplify the following formula.

$$F = \Big((A \wedge B) \to C\Big) \wedge \Big((A \wedge \neg C) \to \neg B\Big) \wedge (A \to B)$$

Solution 10.

$$\begin{aligned}
F &\equiv (\neg(A \wedge B) \vee C) \wedge (\neg(A \wedge \neg C) \vee \neg B) \wedge (\neg A \vee B) && \text{(Implication)} \\
&\equiv (\neg A \vee \neg B \vee C) \wedge (\neg A \vee C \vee \neg B) \wedge (\neg A \vee B) && \text{(De Morgan, Double Negation)} \\
&\equiv (\neg A \vee \neg B \vee C) \wedge (\neg A \vee B) && \text{(Idempotence)} \\
&\equiv \neg A \vee ((\neg B \vee C) \wedge B) && \text{(Distributivity)} \\
&\equiv \neg A \vee ((\neg B \wedge B) \vee (C \wedge B)) && \text{(Distributivity)} \\
&\equiv \neg A \vee (\bot \vee (B \wedge C)) && (F \wedge \neg F \equiv \bot, \text{ Commutativity}) \\
&\equiv \neg A \vee (B \wedge C) && (\bot \vee F \equiv F) \\
&\equiv A \to (B \wedge C) && \text{(Implication)}
\end{aligned}$$

Exercise 11.
Simplify the following formula such that it contains at most one $\neg$ and at most one $\wedge$.

$$F = \Big((A \wedge B) \vee (A \wedge \neg B) \vee (\neg A \wedge B)\Big) \to \neg(A \leftrightarrow B)$$

Solution 11.

$$\begin{aligned}
F &\equiv \neg\Big((A \wedge B) \vee (A \wedge \neg B) \vee (\neg A \wedge B)\Big) \vee \neg\Big((A \wedge B) \vee (\neg A \wedge \neg B)\Big) && \text{(Implication, Equivalence)}\\
&\equiv \neg\Big((A \wedge (B \vee \neg B)) \vee (\neg A \wedge B)\Big) \vee \neg\Big((A \wedge B) \vee (\neg A \wedge \neg B)\Big) && \text{(Distributivity)}\\
&\equiv \neg\Big((A \wedge \top) \vee (\neg A \wedge B)\Big) \vee \neg\Big((A \wedge B) \vee (\neg A \wedge \neg B)\Big) && (F \vee \neg F \equiv \top)\\
&\equiv \neg(A \vee (\neg A \wedge B)) \vee \neg\Big((A \wedge B) \vee (\neg A \wedge \neg B)\Big) && (F \wedge \top \equiv F)\\
&\equiv \neg\Big((A \vee \neg A) \wedge (A \vee B)\Big) \vee \neg\Big((A \wedge B) \vee (\neg A \wedge \neg B)\Big) && \text{(Distributivity)}\\
&\equiv \neg(\top \wedge (A \vee B)) \vee \neg\Big((A \wedge B) \vee (\neg A \wedge \neg B)\Big) && (F \vee \neg F \equiv \top)\\
&\equiv \neg(A \vee B) \vee \neg\Big((A \wedge B) \vee (\neg A \wedge \neg B)\Big) && (F \wedge \top \equiv F, \text{ Comm.})\\
&\equiv (\neg A \wedge \neg B) \vee ((\neg A \vee \neg B) \wedge (A \vee B)) && \text{(De Morgan, DNE)}\\
&\equiv (\neg A \wedge \neg B) \vee (\neg A \wedge A) \vee (\neg A \wedge B) \vee (A \wedge \neg B) \vee (B \wedge \neg B) && \text{(Distributivity, Comm.)}\\
&\equiv (\neg A \wedge \neg B) \vee \bot \vee (\neg A \wedge B) \vee (A \wedge \neg B) \vee \bot && (F \wedge \neg F \equiv \bot)\\
&\equiv (\neg A \wedge \neg B) \vee (\neg A \wedge B) \vee (A \wedge \neg B) && (F \vee \bot \equiv F)\\
&\equiv \Big(\neg A \wedge (\neg B \vee B)\Big) \vee (A \wedge \neg B) && \text{(Distributivity)}\\
&\equiv (\neg A \wedge \top) \vee (A \wedge \neg B) && (F \vee \neg F \equiv \top)\\
&\equiv \neg A \vee (A \wedge \neg B) && (F \wedge \top \equiv F)\\
&\equiv (\neg A \vee A) \wedge (\neg A \vee \neg B) && \text{(Distributivity)}\\
&\equiv \top \wedge (\neg A \vee \neg B) && (F \vee \neg F \equiv \top, \text{ Comm.})\\
&\equiv \neg A \vee \neg B && (F \wedge \top \equiv F, \text{ Comm.})\\
&\equiv \neg(A \wedge B) && \text{(De Morgan)}
\end{aligned}$$

Exercise 12.
Simplify the following formula to an equivalent expression containing only $\vee$ operators between propositional variables.

$$F = \neg\Big((P \to Q) \wedge (Q \to R) \wedge (R \to P)\Big) \vee (P \wedge Q \wedge R)$$

Solution 12.

$$\begin{aligned}
F &\equiv \neg\Big((\neg P \vee Q) \wedge (\neg Q \vee R) \wedge (\neg R \vee P)\Big) \vee (P \wedge Q \wedge R) && \text{(Implication)}\\
&\equiv (P \wedge \neg Q) \vee (Q \wedge \neg R) \vee (R \wedge \neg P) \vee (P \wedge Q \wedge R) && \text{(De Morgan)}\\
&\equiv (P \wedge \neg Q) \vee (Q \wedge \neg R) \vee \Big(R \wedge (\neg P \vee (P \wedge Q))\Big) && \text{(Distributivity)}\\
&\equiv (P \wedge \neg Q) \vee (Q \wedge \neg R) \vee \Big(R \wedge ((\neg P \vee P) \wedge (\neg P \vee Q))\Big) && \text{(Distributivity)}\\
&\equiv (P \wedge \neg Q) \vee (Q \wedge \neg R) \vee \Big(R \wedge (\neg P \vee Q)\Big) && (F \vee \neg F \equiv \top,\ \top \wedge F \equiv F)\\
&\equiv (P \wedge \neg Q) \vee (Q \wedge \neg R) \vee (R \wedge \neg P) \vee (R \wedge Q) && \text{(Distributivity)}\\
&\equiv (P \wedge \neg Q) \vee \Big(Q \wedge (\neg R \vee R)\Big) \vee (R \wedge \neg P) && \text{(Distributivity)}\\
&\equiv (P \wedge \neg Q) \vee (Q \wedge \top) \vee (R \wedge \neg P) && (F \vee \neg F \equiv \top)\\
&\equiv (P \wedge \neg Q) \vee Q \vee (R \wedge \neg P) && (F \wedge \top \equiv F)\\
&\equiv ((P \vee Q) \wedge (\neg Q \vee Q)) \vee (R \wedge \neg P) && \text{(Distributivity)}\\
&\equiv (P \vee Q) \vee (R \wedge \neg P) && (\neg F \vee F \equiv \top,\ F \wedge \top \equiv F)\\
&\equiv (P \vee Q \vee R) \wedge (P \vee Q \vee \neg P) && \text{(Distributivity)}\\
&\equiv (P \vee Q \vee R) \wedge \top && (F \vee \neg F \equiv \top)\\
&\equiv P \vee Q \vee R && (F \wedge \top \equiv F)
\end{aligned}$$

Exercise 13.
Simplify the following formula.

$$F = \Big((A \wedge B) \to C\Big) \to \Big(A \to (B \to C)\Big)$$

Solution 13.

$$\begin{aligned}
F &\equiv \neg\Big(\neg(A \wedge B) \vee C\Big) \vee \Big(\neg A \vee (\neg B \vee C)\Big) && \text{(Implication)}\\
&\equiv \neg(\neg A \vee \neg B \vee C) \vee (\neg A \vee \neg B \vee C) && \text{(De Morgan, Associativity)}\\
&\equiv \top && (F \vee \neg F \equiv \top)
\end{aligned}$$

Exercise 14.
Simplify the following formula such that it contains only $\wedge$ and $\neg$ operators.

$$F = \neg\Big((P \vee Q) \wedge (P \to R) \wedge (Q \to R)\Big) \wedge R$$

Solution 14.

$$\begin{aligned}
F &\equiv \neg\Big((P \vee Q) \wedge (\neg P \vee R) \wedge (\neg Q \vee R)\Big) \wedge R && \text{(Implication)}\\
&\equiv \neg\Big((P \vee Q) \wedge ((\neg P \wedge \neg Q) \vee R)\Big) \wedge R && \text{(Distributivity)}\\
&\equiv \neg\Big((P \vee Q) \wedge (\neg(P \vee Q) \vee R)\Big) \wedge R && \text{(De Morgan)}\\
&\equiv \neg\Big(((P \vee Q) \wedge \neg(P \vee Q)) \vee ((P \vee Q) \wedge R)\Big) \wedge R && \text{(Distributivity)}\\
&\equiv \neg\Big(\bot \vee ((P \vee Q) \wedge R)\Big) \wedge R && (F \wedge \neg F \equiv \bot)\\
&\equiv \neg\Big((P \vee Q) \wedge R\Big) \wedge R && (\bot \vee F \equiv F)\\
&\equiv \Big(\neg(P \vee Q) \vee \neg R\Big) \wedge R && \text{(De Morgan)}\\
&\equiv \Big(\neg(P \vee Q) \wedge R\Big) \vee (\neg R \wedge R) && \text{(Distributivity)}\\
&\equiv (\neg P \wedge \neg Q \wedge R) \vee \bot && \text{(De Morgan, } F \wedge \neg F \equiv \bot)\\
&\equiv \neg P \wedge \neg Q \wedge R && (\bot \vee F \equiv F)
\end{aligned}$$

Exercise 15.
Simplify the following formula.

$$F = \Big((A \to B) \wedge (B \to C)\Big) \wedge \neg(A \to C)$$

Solution 15.

$$\begin{aligned}
F &\equiv (\neg A \vee B) \wedge (\neg B \vee C) \wedge \neg(\neg A \vee C) && \text{(Implication)}\\
&\equiv (\neg A \vee B) \wedge (\neg B \vee C) \wedge (A \wedge \neg C) && \text{(De Morgan, Double Negation)}\\
&\equiv \Big((\neg A \vee B) \wedge A\Big) \wedge \Big((\neg B \vee C) \wedge \neg C\Big) && \text{(Commutativity, Associativity)}\\
&\equiv \Big((\neg A \wedge A) \vee (B \wedge A)\Big) \wedge \Big((\neg B \wedge \neg C) \vee (C \wedge \neg C)\Big) && \text{(Distributivity)}\\
&\equiv (\bot \vee (A \wedge B)) \wedge ((\neg B \wedge \neg C) \vee \bot) && (F \wedge \neg F \equiv \bot, \text{ Commutativity})\\
&\equiv (A \wedge B) \wedge (\neg B \wedge \neg C) && (\bot \vee F \equiv F)\\
&\equiv A \wedge (B \wedge \neg B) \wedge \neg C && \text{(Associativity, Commutativity)}\\
&\equiv A \wedge \bot \wedge \neg C && (F \wedge \neg F \equiv \bot)\\
&\equiv \bot && (\bot \wedge F \equiv \bot)
\end{aligned}$$